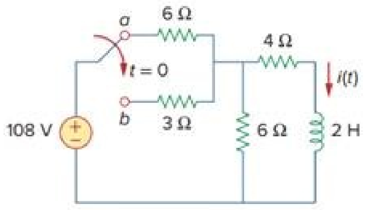
Chapter 7, Problem 78P

Problem

The switch in Fig. 7.142 moves from position *a* to *b* at *t* = 0. Use *PSpice* or *MultiSim* to find *i*(*t*) for *t* > 0.

Fig. 7.142:



Step-by-step solution

Step 1 of 3

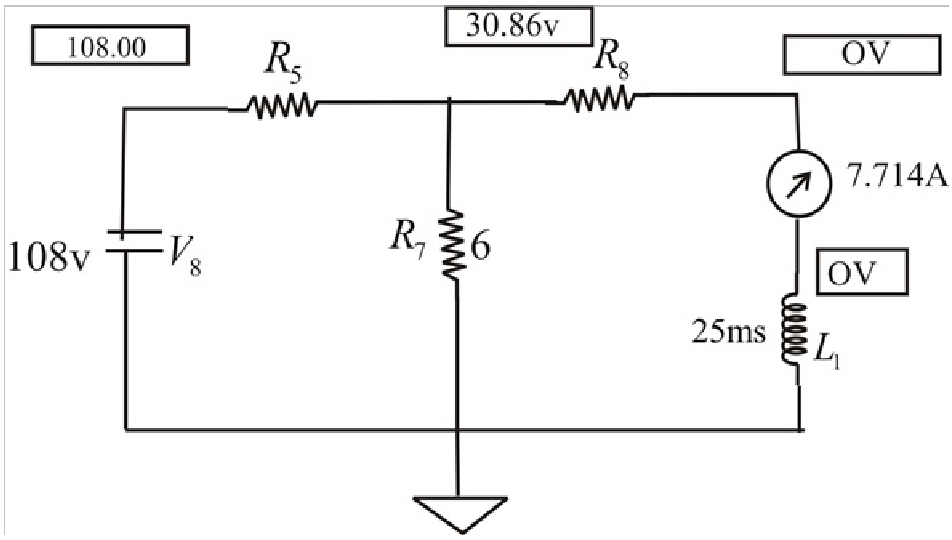
(A) When the switch is in position(a), Schematic is shown below.

We insert iprobe to display i, after simulation, we obtain,

i(0) = 7.714 A.

From the display Iprobe.

Step 2 of 3



Step 3 of 3

(B) When the switch is in position(b), the schematic is as shown below. For Inductor L1, we let L.C = 7.714, by clicking Analysis/Setup/Transient. We let print setup print step = 25 m S.

Final step = 2 S.

After simulation, click Trace/Add in the probe menu & display I(L1), as shown.

Note i(∞) = 12 A, which is correct.